Reference Material

Study of DS-RX1 System Realization (Single Printer)

Aug. 31, 2017
Dai Nippon Printing Co., Ltd.

The copyrights for this document are the property of the holders of rights. Reproduction of any or all of the contents is prohibited.

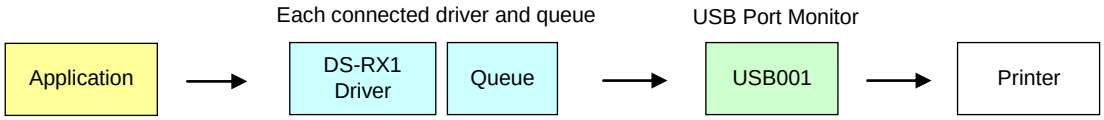
The contents of this document are subject to change without prior notice.

Company and product names used in this document are trademarks or registered trademarks of their respective companies.

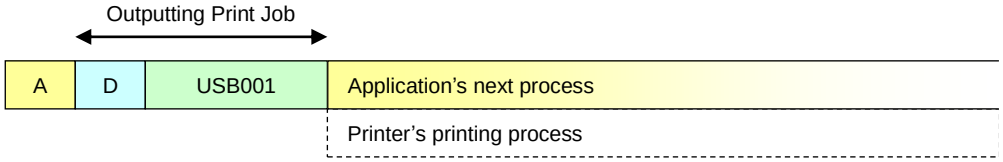
1. About the Data Flow on Windows

In the printer driver settings, there is a Spool On/Off setting. The data flow when using this setting is shown below.

1-1. Operation with Spool Off

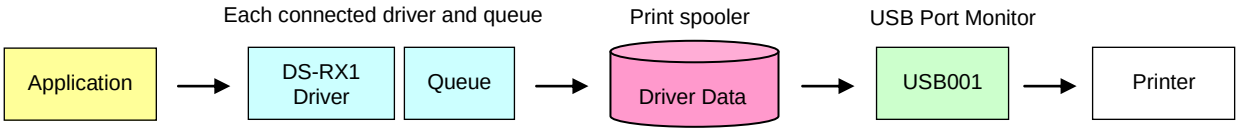


When the application outputs data to the driver, the driver can't release the print job until the data is output to the USB port. Therefore, the application waits until data output to the USB port is complete before performing the next process.

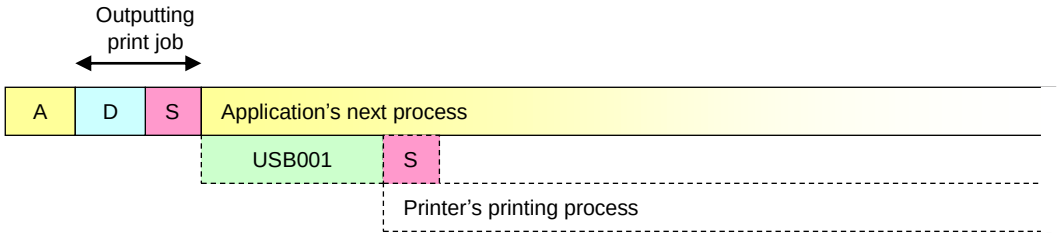


A	Application section: data resizing, status processing (API), sheet no. control, color management, color adjust, etc
D	Driver processing: color management, color adjust, printer command processing, etc.
USB	USB data transferring, data send/receive management

1-2. Operation with Spool On



When the application outputs the data to the driver, the driver outputs the data to the print spooler and releases the print job. At this point, the application can begin the next process without waiting for the data to be output to the USB port. When the print spooler takes in the data, USB Port Monitor control is performed, and the data output to the port begins.



A	Application section: data resizing, status processing (API), sheet no. control, color management, color adjust, etc
D	Driver processing: color management, color adjust, printer command processing, etc.
S	File writing to and deletion from the print spooler
USB	USB data transferring, data send/receive management

[About status check]

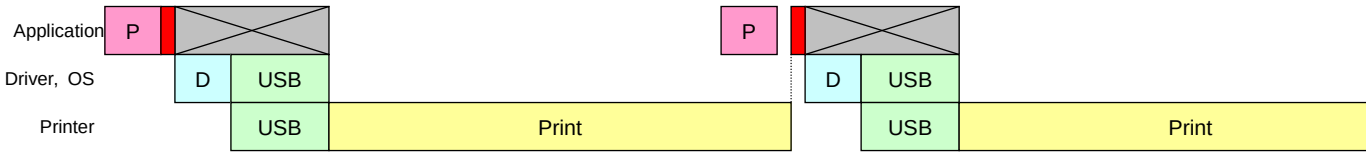
With Spool ON, the application can perform the next process when the data output to the spooler is complete, but until the data output to the port is complete, status confirmation for the printer cannot be done.

2. Relation between printer buffer operation (single/double) and Spool ON/OFF

The relationship between printer buffer operation (single/double) and Spool ON/OFF is shown below.

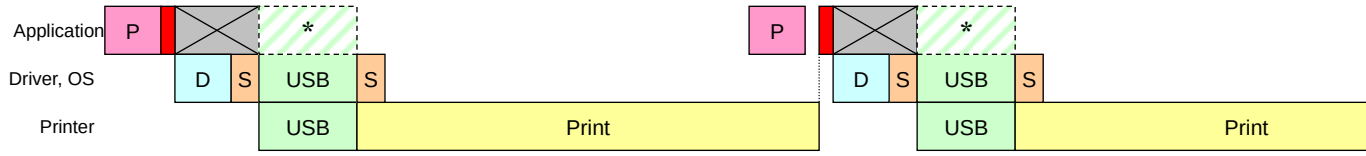
(1) Single buffer, Spool OFF

Because it is Spool OFF, it waits for the data to be sent. Moreover, it waits for printing to be complete before sending the next data.



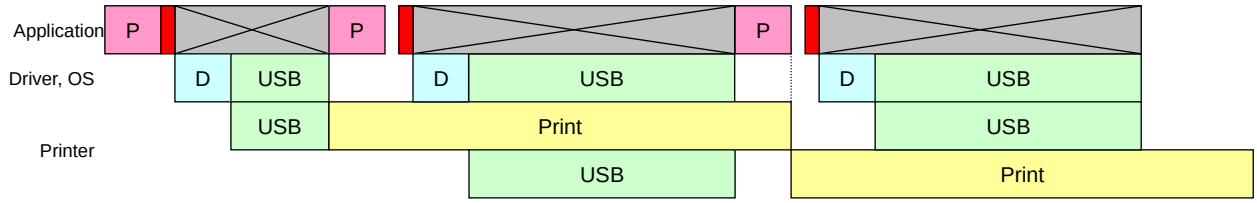
(2) Single buffer, Spool ON

By setting Spool ON, the application is freed from the print processing earlier, but with the spool processing time, the total processing time is slightly longer than with Spool OFF.



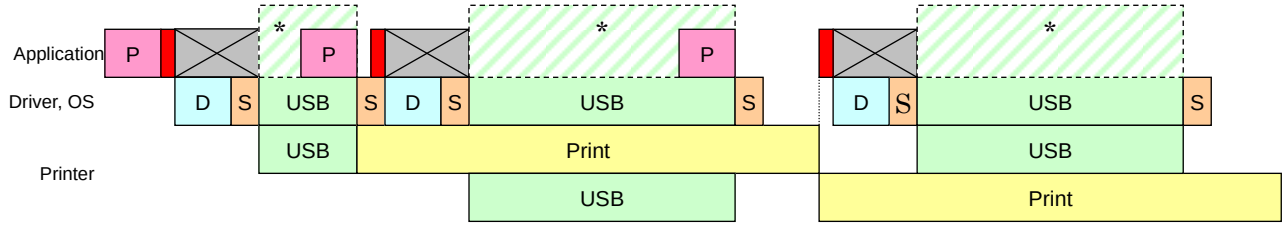
(3) Double buffer, Spool OFF

Using a double buffer, while the first image is being printed, data for the second can be sent, so processing time is that much shorter. However, with Spool OFF, the application cannot perform any other processes while the print job is being sent to the printer.



(4) Double buffer, Spool ON

Double buffer processing allows data to be sent while printing, and with Spool ON, the application can perform other processes (like preparing the next print data), except for printer status confirmation, even while the print job is being sent to the printer.



P	Print data prep (data resize, color management, color adjust)	D	Driver processes (color management, printer commands etc.)
	Start print process	S	File writing to and deletion from the print spooler
	Outputting print job (Application can't perform other processes)	USB	OS, printer USB communication process
*	Printer status confirmation not possible	Print	Printer's printing process

3. System Composition

- Connected printers : 1 Unit
- Printer operating mode : Double buffer operation
- Spool setting : Spool ON



Host DS-RX1

[Control Method]

The general process flow is that, when there is a print request, it checks with the printer to see if data can be sent, then sends the data. The sequence is as follows:

(1) When there is a print request, checks for print job at the printer (Windows API EnumJobs). If there is a print job, repeats print job check.
(When there is a print job, while the Status API tries to get the printer status, the application waits until there is no print job and it can get the status. In order to send data effectively, when there is a print job, it repeats a print job quantity check until there are no more print jobs.)

(2) Performs error check. If there is an error, runs error recovery process (See "4. Error Processing")

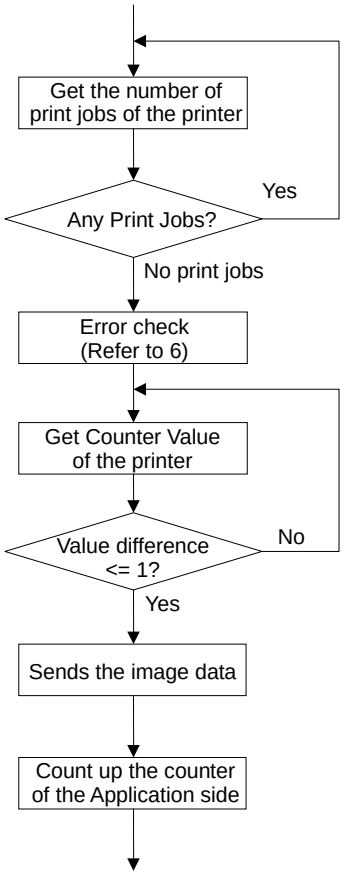
(3) When there are no print jobs or errors, it gets the printer count number (Status API GetCounterL).
It compares the application and printer count values to check for free buffers.

Count value difference = 0 --> 2 free buffers
Count value difference = 1(*1) --> 1 free buffer
Count value difference = 2(*2) --> 0 free buffers

(4) If there is a free buffer, sends print data to printer.

(5) After sending print data, counts up the application side count value.

(6) If there are no free buffers, repeats free buffer check.



(*1) When printing 6x8(A5), 5x7(2L), 6x6(*4), 5x5(*4) or the dual image of 6x4 or 5x3.5(*3) size, the counter value difference will be 2.
(*2) When printing 6x8(A5), 5x7(2L), 6x6(*4), 5x5(*4) or the dual image of 6x4 or 5x3.5(*3) size, the counter value difference will be 4.

*3. Effective at firmware Ver 1.20 or later.
*4. Effective at firmware Ver 2.06 or later.

4. Error processing

When an error occurs, the printer count value is gotten as follows, and based on that value, it is determined how far printing proceeded. Then, once the error has been solved, data for the unprinted images is sent again.

(1) Checks for a print job, and if there is a print job, does not get the printer status.
(2) If there is no print job, gets printer status, and checks for errors.
(3) If there is an error, gets the printer count value.
(4) Compares the printer and application count values.
(5) If the difference in count values is 2(*2), reprocesses the previous 2 data for printing (sends the data to the printer after the error has been solved). If the difference in count values is 1(*1), reprocesses the previous 1 data for printing.

[Remarks]

- If there is a print job for too long, there may be a communication break, so error processing should take place.
- If there is no job, and for some reason there is a communication break, about 1 minute after DS-RX1 Printer Status API tries to get the status, a fail code will be returned.(During this time, the application will be waiting for status) When the program checks the status, it checks the time from start to finish, and if it is taking a long time, error processing is necessary.

